

COUNTEREXAMPLES TO MAKEEV'S REGULAR-SIMPLEX CONJECTURE IN DIMENSIONS FOUR AND FIVE

CHATGPT

ABSTRACT. For $d = 4$ and $d = 5$ we give an explicit smooth odd function on S^{d-1} which does not agree with any linear functional on any rotated normalized A_d root system. In the decomposition $\mathbb{R}^d = \mathbb{C}^2 \oplus \mathbb{R}^{d-4}$, for every sufficiently small nonzero real ε one may take

$$f_\varepsilon(z_1, z_2, x) = \operatorname{Re}(z_1^2 \bar{z}_2) + \varepsilon |z_1|^2 \operatorname{Im}(z_1^2 \bar{z}_2).$$

The two dimensions share the exact-edge-label reduction, a cubic identity forced by a circle symmetry, and the compactness argument. Algebraic transversality is proved directly in dimension four and by an exact rational calculation in dimension five. The latter uses a symmetric multiplication tensor: after linear elimination it is a unit-ideal test involving 55 quadratics and one cubic in 21 variables. The accompanying standard-library Python generator implements the tensor test for every fixed $d \geq 4$, emits the dimension-four and dimension-five Magma certificates, and reproduces the successfully verified dimension-five file byte for byte. The complete direct dimension-four calculation and the generator source are included as appendices.

1. INTRODUCTION

Kuperberg showed that a constant-width formulation of Makeev's universal-cover conjecture is equivalent to asking whether every continuous odd function on S^{d-1} agrees with a linear function on a rotated A_d root system. He proved the assertion for $d = 3$ and showed that the homological obstruction used there vanishes in higher dimensions [2]. The following theorem gives counterexamples in the next two dimensions.

Put $N = d + 1$ and

$$V_d = \mathbf{1}^\perp \subset \mathbb{R}^N, \quad q_i = e_i - \frac{1}{N} \mathbf{1}, \quad u_{ij} = \frac{q_i - q_j}{\sqrt{2}} \quad (0 \leq i < j \leq d).$$

Thus the u_{ij} are the normalized roots of type A_d .

Theorem 1.1 (Counterexamples in dimensions four and five). *For each $d \in \{4, 5\}$ there exists $\varepsilon_0(d) > 0$ such that, whenever $0 < |\varepsilon| < \varepsilon_0(d)$, the smooth odd function*

$$f_\varepsilon(z_1, z_2, x) = \operatorname{Re}(z_1^2 \bar{z}_2) + \varepsilon |z_1|^2 \operatorname{Im}(z_1^2 \bar{z}_2), \quad (z_1, z_2, x) \in S^{d-1} \subset \mathbb{C}^2 \oplus \mathbb{R}^{d-4}, \quad (1)$$

has no isometry $g : V_d \rightarrow \mathbb{R}^d$ and $b \in \mathbb{R}^d$ satisfying

$$f_\varepsilon(gu_{ij}) = b \cdot gu_{ij} \quad (0 \leq i < j \leq d).$$

Consequently Makeev's conjecture fails in dimensions four and five.

A transposition of two simplex vertices induces an orientation-reversing symmetry of the root system. Hence the formulations with $O(d)$ and $SO(d)$ are equivalent. We freely use either one below.

The proof has one dimension-dependent step. We first give the common geometric and variational reductions. We then encode the remaining algebraic assertion by a multiplication tensor. In dimension five the resulting rational ideal was proved to be the unit ideal by Magma. In dimension

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This manuscript and its accompanying certificate generator are AI-generated. The dimension-five Magma certificate was independently run by the human verifier.

four the complete direct proof from the original argument is included in [appendix A](#); the generated tensor certificate provides an independent computational audit.

2. REGULAR-SIMPLEX LABELS

The simplex vectors satisfy

$$\sum_{i=0}^d q_i = 0, \quad \langle q_i, q_j \rangle = \delta_{ij} - \frac{1}{N}, \quad \sum_{i=0}^d q_i q_i^T = I_{V_d}. \quad (2)$$

In particular $\|q_i - q_j\|^2 = 2$, so every u_{ij} lies on the unit sphere.

For an odd function $F : S^{d-1} \rightarrow \mathbb{R}$ and an isometry $g : V_d \rightarrow \mathbb{R}^d$, define the skew matrix

$$Y_F(g)_{ij} = F(gu_{ij}), \quad Y_F(g)_{ji} = -Y_F(g)_{ij}, \quad (3)$$

and put

$$\Pi = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T.$$

Lemma 2.1 (Exact edge labels). *For a real skew $N \times N$ matrix Y , the following are equivalent:*

- (1) $Y_{ij} = c \cdot u_{ij}$ for some $c \in V_d$;
- (2) $Y_{ij} = \rho_i - \rho_j$ for some $\rho_0, \dots, \rho_d \in \mathbb{R}$;
- (3) $\Pi Y \Pi = 0$.

Proof. If $Y_{ij} = c \cdot u_{ij}$, take $\rho_i = (c \cdot q_i)/\sqrt{2}$. Conversely, subtract the mean of the ρ_i and set $c = \sqrt{2} \sum_i \rho_i q_i$. The Gram identity (2) gives $c \cdot q_j = \sqrt{2} \rho_j$.

If $Y_{ij} = \rho_i - \rho_j$, then $Y = \rho \mathbf{1}^T - \mathbf{1} \rho^T$, so $\Pi Y \Pi = 0$. Conversely, with $J_0 = N^{-1} \mathbf{1}\mathbf{1}^T$, skew-symmetry gives $J_0 Y J_0 = 0$, and $\Pi Y \Pi = 0$ gives $Y = J_0 Y + Y J_0$. Taking $\rho_i = N^{-1} \sum_j Y_{ij}$ proves $Y_{ij} = \rho_i - \rho_j$. $\square$

We call these matrices *exact*. If X is skew and $X \mathbf{1} = 0$, then

$$\sum_{i < j} X_{ij}(\rho_i - \rho_j) = 0. \quad (4)$$

This is the cut/cycle decomposition of complete-graph edge flows [1]. By [lemma 2.1](#), the equality in [theorem 1.1](#) occurs precisely when $Y_{f_\varepsilon}(g)$ is exact.

3. THE COMMON CUBIC ANNIHILATOR

On $\mathbb{C}^2 \oplus \mathbb{R}^{d-4}$ put

$$W(z) = z_1^2 \bar{z}_2, \quad Q = \operatorname{Re} W, \quad R = \operatorname{Im} W, \quad H = |z_1|^2 R. \quad (5)$$

Consider the circle action and its infinitesimal generator

$$h_\theta(z_1, z_2, x) = (e^{i\theta} z_1, e^{2i\theta} z_2, x), \quad J(z_1, z_2, x) = (iz_1, 2iz_2, 0). \quad (6)$$

Both Q and H are invariant under this action.

Write $p_i = gq_i$ and define

$$X(g)_{ij} = \langle p_i, J p_j \rangle_{\mathbb{R}}, \quad \Lambda_g(Y) = \sum_{i < j} X(g)_{ij} Y_{ij}. \quad (7)$$

The matrix $X(g)$ is skew and $\sum_j X(g)_{ij} = \langle p_i, J \sum_j p_j \rangle_{\mathbb{R}} = 0$. Thus (4) shows that Λ_g annihilates every exact label matrix.

Let

$$P_3(v) = \sum_{i=0}^d \langle q_i, v \rangle_{\mathbb{R}}^3.$$

For a homogeneous cubic C , write T_C for its symmetric trilinear polarization and use the invariant inner product

$$\langle C, D \rangle = \sum_{\alpha, \beta, \gamma=1}^d T_C(e_\alpha, e_\beta, e_\gamma) T_D(e_\alpha, e_\beta, e_\gamma).$$

For $K \in \mathfrak{so}(d)$, let $(K \cdot C)(x) = \frac{d}{dt} \Big|_{t=0} C(e^{-tK}x)$.

Lemma 3.1 (Cubic edge identity). *For every homogeneous cubic C and $K \in \mathfrak{so}(d)$,*

$$\langle C, K \cdot P_3 \rangle = 2\sqrt{2} \sum_{i < j} \langle q_i, K q_j \rangle_{\mathbb{R}} C(u_{ij}). \quad (8)$$

Proof. The Parseval identity in (2) gives

$$\langle C, K \cdot P_3 \rangle = 3 \sum_i T_C(q_i, q_i, K q_i).$$

Since $C(u_{ij}) = C(q_i - q_j)/(2\sqrt{2})$, the right side of (8) is

$$\sum_{i < j} \langle q_i, K q_j \rangle_{\mathbb{R}} C(q_i - q_j).$$

The pure terms in its cubic expansion vanish because the row sums of $(\langle q_i, K q_j \rangle)_{ij}$ vanish. The mixed terms combine to

$$-3 \sum_{i,j} \langle q_i, K q_j \rangle_{\mathbb{R}} T_C(q_i, q_i, q_j).$$

Finally $K q_i = -\sum_j \langle q_i, K q_j \rangle_{\mathbb{R}} q_j$, again by Parseval, and the last display is $3 \sum_i T_C(q_i, q_i, K q_i)$. $\square$

Lemma 3.2 (Global cubic annihilation). *For every $d \geq 4$ and every isometry $g : V_d \rightarrow \mathbb{R}^d$,*

$$\Lambda_g(Y_Q(g)) = 0. \quad (9)$$

Proof. Take $K = g^{-1}Jg$ and $C = Q \circ g$ in lemma 3.1. The orthogonal action on cubic tensors is infinitesimally skew-adjoint, while $K \cdot (Q \circ g) = (J \cdot Q) \circ g = 0$ by circle invariance. Hence $\langle Q \circ g, K \cdot P_3 \rangle = 0$, and the edge identity gives the claim. $\square$

The remaining statement is algebraic.

Proposition 3.3 (Algebraic transversality). *For $d = 4$ and $d = 5$,*

$$Y_Q(g) \text{ exact} \implies \Lambda_g(Y_H(g)) \neq 0. \quad (10)$$

The direct proof for $d = 4$ is [appendix A](#). The common algebraic encoding and the exact $d = 5$ computation occupy the next two sections.

4. THE CUBIC EXACTNESS LOCUS

Write

$$p_i = (a_i, b_i, c_i^{(4)}, \dots, c_i^{(d-1)}) \in \mathbb{C}^2 \oplus \mathbb{R}^{d-4}.$$

Thus $a, b \in \mathbb{C}^N$, while each $c^{(r)} \in \mathbb{R}^N$. The real frame identities say that

$$\mathbf{1}, a, b, \bar{a}, \bar{b}, c^{(4)}, \dots, c^{(d-1)}$$

are mutually orthogonal in $\mathbb{C}^N$. Their squared norms are respectively $N, 2, 2, 2, 2, 1, \dots, 1$.

At $gu_{ij} = (p_i - p_j)/\sqrt{2}$, the complex cubic has, apart from the common factor $1/(2\sqrt{2})$, the edge label

$$\tilde{Y}_{ij} = (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

For $(u \wedge v)_{ij} = u_i v_j - v_i u_j$, expansion modulo exact matrices gives

$$\omega := \Pi \tilde{Y} \Pi = -a^2 \wedge \bar{b} + 2a \wedge (a\bar{b}). \quad (11)$$

The Q -labels are exact precisely when $\operatorname{Re} \omega = 0$.

Define

$$\begin{aligned} s &= \frac{1}{2} \sum_i |a_i|^2 a_i, & m &= \sum_i a_i^2 \bar{b}_i, & n &= \sum_i a_i^3, & \ell &= \frac{1}{2} \sum_i a_i^2 b_i, \\ \lambda &= \frac{1}{2} \sum_i |a_i|^2 \bar{b}_i, & u &= \frac{1}{2} \sum_i a_i \bar{b}_i^2, & \delta &= \frac{1}{2} \sum_i a_i |b_i|^2, \\ r_r &= \sum_i c_i^{(r)} a_i^2, & w_r &= \sum_i c_i^{(r)} a_i \bar{b}_i & (4 \leq r < d). \end{aligned}$$

Expansion in the displayed orthogonal basis gives

$$a^2 = sa + \frac{m}{2}b + \frac{n}{2}\bar{a} + \ell\bar{b} + \sum_{r=4}^{d-1} r_r c^{(r)}, \quad (12)$$

$$a\bar{b} = \lambda a + ub + \frac{m}{2}\bar{a} + \delta\bar{b} + \sum_{r=4}^{d-1} w_r c^{(r)}. \quad (13)$$

Substitution in (11), followed by comparison with the complex conjugate, proves the next criterion.

Proposition 4.1 (Dimension-independent exactness criterion). *The Q -labels are exact if and only if*

$$m \in \mathbb{R}, \quad 4u = \bar{n}, \quad r_r = w_r = 0 \quad (4 \leq r < d), \quad 2\delta = s. \quad (14)$$

Equivalently, m is real and the two coordinatewise vector identities

$$a^2 = sa + \frac{m}{2}b + \frac{n}{2}\bar{a} + \ell\bar{b}, \quad (15)$$

$$a\bar{b} = \lambda a + \frac{\bar{n}}{4}b + \frac{m}{2}\bar{a} + \frac{s}{2}\bar{b} \quad (16)$$

hold.

Proof. Substituting (12)–(13) into (11) yields

$$\begin{aligned} \omega &= 2ua \wedge b + 2 \sum_r w_r a \wedge c^{(r)} + m a \wedge \bar{a} + (2\delta - s)a \wedge \bar{b} \\ &\quad - \frac{m}{2}b \wedge \bar{b} - \sum_r r_r c^{(r)} \wedge \bar{b} - \frac{n}{2}\bar{a} \wedge \bar{b}. \end{aligned}$$

The wedges of the members of the orthogonal basis are linearly independent. Thus $\omega + \bar{\omega} = 0$ is equivalent to (14). Substitution in (12)–(13) then gives (15)–(16), and the converse follows by reversing the calculation. $\square$

There are $4d - 11$ real linear conditions in (14): five common conditions and four for every extra real coordinate. This linearity becomes particularly useful in tensor coordinates.

5. THE MULTIPLICATION-TENSOR CERTIFICATE

Write $a = x + iy$ and $b = p + iq$, and set

$$e_* = N^{-1/2} \mathbf{1}, \quad v_0 = x, \quad v_1 = y, \quad v_2 = p, \quad v_3 = q, \quad v_r = c^{(r)} \quad (4 \leq r < d).$$

Then $e_*, v_0, \dots, v_{d-1}$ is an orthonormal basis of $\mathbb{R}^N$. Define the fully symmetric tensor

$$T_{ijk} = \sum_{\nu=0}^d v_i(\nu) v_j(\nu) v_k(\nu), \quad 0 \leq i, j, k < d. \quad (17)$$

Coordinatewise multiplication by v_i , in this basis, has the real symmetric matrix

$$M_i = \begin{pmatrix} 0 & N^{-1/2}(\delta_{ij})_{j=0}^{d-1} \\ N^{-1/2}(\delta_{ik})_{k=0}^{d-1} & (T_{ijk})_{j,k=0}^{d-1} \end{pmatrix}. \quad (18)$$

Coordinatewise multiplication gives

$$\sum_{j=0}^{d-1} T_{ijj} = 0, \quad [M_i, M_j] = 0 \quad (0 \leq i < j < d). \quad (19)$$

Lemma 5.1 (Exact tensor model). *A real symmetric tensor T arises from an orthonormal simplex frame by (17) if and only if it satisfies (19).*

Proof. Necessity was just observed. Conversely, use T to define a commutative algebra with unit $\sqrt{N}e_*$ and multiplication table

$$v_i v_j = \frac{\delta_{ij}}{\sqrt{N}} e_* + \sum_{k=0}^{d-1} T_{ijk} v_k.$$

The matrices M_i are its left-multiplication operators. Their commutation, together with commutativity of the product, implies associativity. Hence

$$M_i M_j = \frac{\delta_{ij}}{N} I_N + \sum_{k=0}^{d-1} T_{ijk} M_k. \quad (20)$$

The trace equations give $\text{tr } M_k = 0$, so taking traces in (20) yields $\text{tr}(M_i M_j) = \delta_{ij}$.

The commuting real symmetric matrices are simultaneously orthogonally diagonalizable. Their joint eigenvalue vectors are therefore an orthonormal basis of $\mathbf{1}^\perp \subset \mathbb{R}^N$ (zero trace gives orthogonality to $\mathbf{1}$). Finally,

$$\text{tr}(M_i M_j M_k) = T_{ijk},$$

by (20); in the common eigenbasis this trace is exactly the right side of (17). Thus no real frames are lost and no spurious real frames are introduced. $\square$

Only the lower-right block of each commutator is nontrivial. Its independent entries are

$$\sum_{m=0}^{d-1} (T_{ikm} T_{jml} - T_{jkm} T_{iml}) + \frac{\delta_{ki} \delta_{jl} - \delta_{kj} \delta_{il}}{N} = 0, \quad (21)$$

where $i < j$ and $k < \ell$.

In zero-based indices $0, 1, 2, 3 = x, y, p, q$, the exactness equations of [proposition 4.1](#) are the following rational linear equations:

$$\begin{aligned} 2T_{012} - T_{003} + T_{113} &= 0, \\ 2T_{022} - 2T_{033} + 4T_{123} - T_{000} + 3T_{011} &= 0, \\ 2T_{122} - 2T_{133} - 4T_{023} + 3T_{001} - T_{111} &= 0, \\ T_{00r} - T_{11r} = 2T_{01r} = T_{02r} + T_{13r} = T_{12r} - T_{03r} &= 0 \quad (4 \leq r < d), \\ 2T_{022} + 2T_{033} - T_{000} - T_{011} &= 0, \\ 2T_{122} + 2T_{133} - T_{001} - T_{111} &= 0. \end{aligned} \quad (22)$$

Together with the d trace equations, these are $5d - 11$ independent linear conditions in the $\binom{d+2}{3}$ tensor coordinates. Exact rational row reduction therefore leaves

$$r_d = \binom{d+2}{3} - (5d - 11) \quad (23)$$

free variables.

It remains to express the obstruction. In addition to s, m, n, ℓ, λ from [section 4](#), put

$$v = \sum_i a_i b_i^2, \quad k = \sum_i b_i^3, \quad \rho = \sum_i |b_i|^2 b_i.$$

All eight moments are linear in T :

$$\begin{aligned} s &= \frac{1}{2}(T_{000} + T_{011}) + \frac{i}{2}(T_{001} + T_{111}), \\ m &= T_{002} - T_{112} + 2T_{013}, \\ n &= (T_{000} - 3T_{011}) + i(3T_{001} - T_{111}), \\ \ell &= \frac{1}{2}(T_{002} - T_{112} - 2T_{013}) + \frac{i}{2}(T_{003} - T_{113} + 2T_{012}), \\ \lambda &= \frac{1}{2}(T_{002} + T_{112}) - \frac{i}{2}(T_{003} + T_{113}), \\ v &= (T_{022} - T_{033} - 2T_{123}) + i(T_{122} - T_{133} + 2T_{023}), \\ k &= (T_{222} - 3T_{233}) + i(3T_{223} - T_{333}), \\ \rho &= (T_{222} + T_{233}) + i(T_{223} + T_{333}). \end{aligned}$$

On the linear exactness locus, m is real. Define

$$\begin{aligned} \mathcal{E} = & -\frac{3}{2}m^3 + \frac{3}{4}\bar{n}mn - 12\bar{\rho}\ell\lambda - 16\bar{\lambda}\lambda m - 2\bar{n}\bar{s}\ell - 6\rho\lambda m - 8\bar{s}\lambda n - 8\lambda s^2 - \bar{v}mv \\ & + 12\bar{k}\bar{\lambda}\ell + 14\bar{\ell}\ell m + 16\ell\lambda^2 + 4\bar{\lambda}\bar{v}n + 5\bar{\ell}ns + 6\bar{\lambda}\bar{\rho}m + 6\bar{v}\ell s, \quad E = \operatorname{Re} \mathcal{E}. \end{aligned} \quad (24)$$

Lemma 5.2 (Tensor obstruction identity). *On the cubic exactness locus, in every dimension $d \geq 4$,*

$$E = 8\sqrt{2} \Lambda_g(Y_H(g)). \quad (25)$$

Proof. At an edge, the degree-five factor in H is

$$\frac{1}{4\sqrt{2}}(a_i - a_j)^3(\bar{a}_i - \bar{a}_j)(\bar{b}_i - \bar{b}_j),$$

and

$$X(g) = \frac{i}{2}(\bar{a} \wedge a + 2\bar{b} \wedge b).$$

Expand the edge factor modulo exact matrices, pair its wedges with $X(g)$ using

$$\sum_{i < j} (u \wedge v)_{ij} (x \wedge y)_{ij} = (u \cdot x)(v \cdot y) - (u \cdot y)(v \cdot x),$$

and reduce every coordinatewise product by (15)–(16). Collecting the real part gives exactly (24). This calculation is independent of the number of extra real frame vectors because their coefficients vanish in the two universal identities. Keeping the edge normalization gives the factor $8\sqrt{2}$ in (25). The full uncollected wedge expansion and its specialization to $d = 4$ appear in [appendix A](#); the generator in the next section implements the collected rational expression term by term. $\square$

Thus algebraic transversality follows if the ideal generated by (21) and E becomes the unit ideal after the linear equations have been eliminated. By [lemma 5.1](#), this test loses no real simplex frames. A unit ideal over $\mathbb{C}$ is stronger than needed, since it excludes complex as well as real common zeros.

6. EXACT COMPUTATIONS IN DIMENSIONS FOUR AND FIVE

The accompanying script `tensor_polynomials.py` uses only the Python standard library. For a selected $d \geq 4$ it:

- (1) enumerates the symmetric coordinates T_{ijk} ;
- (2) performs exact rational row reduction on the $5d - 11$ linear equations;
- (3) substitutes the resulting parametrization into (21) and (24);

- (4) clears only nonzero rational scalar factors, deletes zero and duplicate commutators, and emits a self-contained Magma program over $\mathbb{Q}$.

The resulting systems are summarized below.

d	tensor variables	linear equations	free variables	quadratics	cubics
4	20	9	11	21	1
5	35	14	21	55	1

The dimension-four obstruction has 73 monomials after substitution and is $E/4$ up to the generator's harmless normalization. The dimension-five obstruction has 163 monomials and is $9E/4$. Generate and run both files by

```
python3 tensor_polynomials.py --dimension 4 \
--emit-magma makeev_d4_tensor_certificate.m
python3 tensor_polynomials.py --dimension 5 \
--emit-magma makeev_d5_tensor_certificate.m
magma makeev_d4_tensor_certificate.m
magma makeev_d5_tensor_certificate.m
```

Each generated program forms the ideal, enables Magma's modular Gröbner-basis arithmetic, calls `IsProper`, asserts that the answer is false, and prints

Certificate verified: tensor ideal is the unit ideal.

only after that assertion passes.

For $d = 5$, the generated file is byte-for-byte identical to the certificate that was run successfully in approximately 14 seconds. The exact source digests for the accompanying files are

```
fe71e120af7f7c38806701e07042f88bff461c5a19738653d78a0fbe9a7677ff
tensor_polynomials.py
4ae9030dd0b57c9541c7e6861d9d87a956015e009bd5220c2903eb222018d8d7
makeev_d4_tensor_certificate.m
403d831600c2ac4a6d60ba768506ff031eb1d4817eadde04d80e40e3aa14d2e6
makeev_d5_tensor_certificate.m.
```

Therefore the dimension-five part of [proposition 3.3](#) is an exact, unconditional consequence of that unit-ideal computation. The supplied dimension-four Magma file is an independent audit of the direct proof in [appendix A](#); the dimension-four theorem does not depend on an unreported computer run.

The generator accepts every fixed $d \geq 4$. This is a way to formulate higher-dimensional tests, not a claim that their Gröbner-basis computations will terminate quickly. The number r_d of free variables grows cubically. In particular, the older coordinate program `makeev_general_certificate.m`, which is slow already for $d = 4, 5$, is not used anywhere in the proof.

7. COMPLETION OF THE PROOF

Fix $d \in \{4, 5\}$. Suppose the theorem failed for arbitrarily small nonzero ε . Then there would be $\varepsilon_\nu \rightarrow 0$, $\varepsilon_\nu \neq 0$, and isometries $g_\nu : V_d \rightarrow \mathbb{R}^d$ for which $Y_{Q+\varepsilon_\nu H}(g_\nu)$ is exact. Equivalently,

$$\Pi Y_Q(g_\nu) \Pi + \varepsilon_\nu \Pi Y_H(g_\nu) \Pi = 0. \quad (26)$$

After passing to a subsequence, compactness of the orthogonal group gives $g_\nu \rightarrow g$. Letting $\nu \rightarrow \infty$ in (26) shows that $Y_Q(g)$ is exact.

For every ν , the functional Λ_{g_ν} annihilates exact labels; also $\Lambda_{g_\nu}(Y_Q(g_\nu)) = 0$ by [lemma 3.2](#). Since $\varepsilon_\nu \neq 0$, exactness of the $Q + \varepsilon_\nu H$ labels therefore gives

$$\Lambda_{g_\nu}(Y_H(g_\nu)) = 0.$$

Passing to the limit yields $\Lambda_g(Y_H(g)) = 0$. This contradicts [proposition 3.3](#), because $Y_Q(g)$ is exact. Hence an $\varepsilon_0(d) > 0$ exists. The polynomial in (1) is smooth and odd, completing the proof.

APPENDIX A. DIRECT PROOF OF ALGEBRAIC TRANSVERSALITY IN DIMENSION FOUR

This appendix reproduces the complete algebraic appendix of the original dimension-four argument. It proves the $d = 4$ assertion of [proposition 3.3](#) without computer assistance.

Fix an isometry $g : V_4 \rightarrow \mathbb{R}^4 \cong \mathbb{C}^2$ for which the Q -labels are exact, write

$$p_i = gq_i = (a_i, b_i) \in \mathbb{C}^2, \quad a = (a_0, \dots, a_4), \quad b = (b_0, \dots, b_4) \in \mathbb{C}^5,$$

and use coordinatewise multiplication of vectors in $\mathbb{C}^5$. The Hermitian inner product is conjugate-linear in the first variable:

$$\langle x, y \rangle_{\mathbb{C}^5} = \sum_i \bar{x}_i y_i.$$

A.1. Frame identities. The 5×4 real matrix whose rows are the p_i has orthonormal columns, all perpendicular to $\mathbf{1}$. Therefore

$$\sum_i a_i = \sum_i b_i = 0, \tag{27}$$

$$\sum_i a_i^2 = \sum_i b_i^2 = \sum_i a_i b_i = \sum_i a_i \bar{b}_i = 0, \tag{28}$$

$$\sum_i |a_i|^2 = \sum_i |b_i|^2 = 2, \tag{29}$$

$$|a_i|^2 + |b_i|^2 = \frac{4}{5}. \tag{30}$$

Consequently

$$\mathbf{1}, a, b, \bar{a}, \bar{b}$$

are mutually orthogonal in $\mathbb{C}^5$, with squared norms 5, 2, 2, 2, 2. Every $x \in \mathbb{C}^5$ has the expansion

$$x = \frac{\sum_i x_i}{5} \mathbf{1} + \frac{1}{2} \left(\langle a, x \rangle a + \langle b, x \rangle b + \langle \bar{a}, x \rangle \bar{a} + \langle \bar{b}, x \rangle \bar{b} \right). \tag{31}$$

For $u, v \in \mathbb{C}^5$, write

$$(u \wedge v)_{ij} = u_i v_j - v_i u_j.$$

A.2. Cycle equations for the cubic. At the root $gu_{ij} = (p_i - p_j)/\sqrt{2}$,

$$W(gu_{ij}) = \frac{1}{2\sqrt{2}} (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

Ignoring the harmless factor $1/(2\sqrt{2})$, let

$$\tilde{Y}_{ij} = (a_i - a_j)^2 (\bar{b}_i - \bar{b}_j).$$

Modulo exact matrices, expansion gives

$$\Pi \tilde{Y} \Pi \equiv -a^2 \wedge \bar{b} + 2a \wedge (a\bar{b}). \tag{32}$$

Indeed, the omitted term is $a_i^2 \bar{b}_i - a_j^2 \bar{b}_j$.

Define

$$s = \frac{1}{2} \sum_i |a_i|^2 a_i, \quad \lambda = \frac{1}{2} \sum_i |a_i|^2 \bar{b}_i, \tag{33}$$

$$m = \sum_i a_i^2 \bar{b}_i, \quad n = \sum_i a_i^3, \quad u = \frac{1}{2} \sum_i a_i \bar{b}_i^2, \quad \ell = \frac{1}{2} \sum_i a_i^2 b_i. \tag{34}$$

Using (31),

$$a^2 = sa + \frac{m}{2}b + \frac{n}{2}\bar{a} + \ell\bar{b}, \quad (35)$$

$$a\bar{b} = \lambda a + ub + \frac{m}{2}\bar{a} - s\bar{b}. \quad (36)$$

Substitution into (32) yields

$$\omega := \Pi\tilde{Y}\Pi = 2u a \wedge b + m a \wedge \bar{a} - 3s a \wedge \bar{b} - \frac{m}{2}b \wedge \bar{b} - \frac{n}{2}\bar{a} \wedge \bar{b}. \quad (37)$$

Exactness of the real Q -labels is precisely $\operatorname{Re} \omega = 0$. The six wedges of the complex basis $a, b, \bar{a}, \bar{b}$ are linearly independent. Comparing (37) with its conjugate gives

$$s = 0, \quad m \in \mathbb{R}, \quad 4u = \bar{n}. \quad (38)$$

Replacing g by $h_\theta g$ changes neither the values of Q and H nor the matrix $X(g)$, while $\lambda \mapsto e^{-2i\theta}\lambda$. We may therefore assume

$$\lambda = t \geq 0. \quad (39)$$

In addition define

$$v = \sum_i a_i b_i^2, \quad k = \sum_i b_i^3. \quad (40)$$

A.3. The coordinatewise multiplication table. Expanding each product in the orthogonal basis $\mathbf{1}, a, b, \bar{a}, \bar{b}$ by (31), and using (30) and (38), gives

$$\begin{aligned} a\bar{a} &= \frac{2}{5}\mathbf{1} + tb + t\bar{b}, & b\bar{b} &= \frac{2}{5}\mathbf{1} - tb - t\bar{b}, \\ a^2 &= \frac{m}{2}b + \frac{n}{2}\bar{a} + \ell\bar{b}, & \bar{a}^2 &= \frac{m}{2}\bar{b} + \frac{\bar{n}}{2}a + \bar{\ell}b, \\ a\bar{b} &= ta + \frac{\bar{n}}{4}b + \frac{m}{2}\bar{a}, & \bar{a}b &= t\bar{a} + \frac{n}{4}\bar{b} + \frac{m}{2}a, \\ ab &= ta + \ell\bar{a} + \frac{v}{2}\bar{b}, & \bar{a}\bar{b} &= t\bar{a} + \bar{\ell}a + \frac{\bar{v}}{2}b, \\ b^2 &= \frac{n}{4}a - tb + \frac{v}{2}\bar{a} + \frac{k}{2}\bar{b}, & \bar{b}^2 &= \frac{\bar{n}}{4}\bar{a} - t\bar{b} + \frac{\bar{v}}{2}a + \frac{\bar{k}}{2}b. \end{aligned} \quad (41)$$

For example, the coefficient of b in $a\bar{a}$ is $\frac{1}{2} \sum_i \bar{b}_i |a_i|^2 = t$, and the other coefficients are obtained in the same way.

Coordinatewise multiplication is associative. Comparing coefficients in (41) gives the following necessary equations:

$$3mn = 4tv, \quad (E1)$$

$$\bar{n}v + 16\ell t + 8mt = 0, \quad (E2)$$

$$-2km + 24\ell t - n^2 = 0, \quad (E3)$$

$$-kt + \ell m + 4t^2 = 0, \quad (E4)$$

$$\bar{n}k + mn + 8tv = 0, \quad (E5)$$

$$20|\ell|^2 + 5m^2 + 5|n|^2 - 40t^2 = 8, \quad (E6)$$

$$16|\ell|^2 - 4m^2 - |n|^2 + 4|v|^2 = 0, \quad (E7)$$

$$20|\ell|^2 + 60t^2 + 5|v|^2 = 8. \quad (E8)$$

More precisely, (E1)–(E3) are the $a, b, \bar{b}$ coefficients of $(a^2)b = a(ab)$; (E4)–(E5) are the a, b coefficients of $(ab)b = a(b^2)$; and (E6), (E7), (E8) are respectively the indicated coefficients of

$$(a^2)\bar{a} = a(a\bar{a}), \quad (ab)\bar{a} = a(b\bar{a}), \quad (ab)\bar{b} = a(b\bar{b}).$$

A.4. Classification of the possible cubic zeros. We extract the consequences of (E1)–(E8) needed below.

Case I: $t = 0$. Equation (E1) gives $mn = 0$. If $m = 0$, then (E3) gives $n = 0$, while (E6)–(E7) become

$$20|\ell|^2 = 8, \quad 16|\ell|^2 + 4|v|^2 = 0,$$

a contradiction. Hence $m \neq 0$ and $n = 0$. Equations (E4) and (E3) give $\ell = k = 0$, and then

$$m^2 = \frac{8}{5}, \quad |v|^2 = m^2. \quad (42)$$

Case II: $t > 0$ and $n = 0$. Equation (E1) gives $v = 0$, and (E2) gives

$$\ell = -\frac{m}{2}. \quad (43)$$

Here $m \neq 0$, since otherwise (41) would give $a^2 = 0$. Equations (E3) and (E4) give $k = -6t$ and $m^2 = 20t^2$. Finally (E8) yields

$$t^2 = \frac{1}{20}, \quad m^2 = 1. \quad (44)$$

Case III: $t > 0$ and $n \neq 0$. First $m \neq 0$, since otherwise (E1), (E2), and (E3) successively give $v = 0$, $\ell = 0$, and $n = 0$. Put

$$q = \frac{m}{t}, \quad h = \frac{|n|^2}{t^2} > 0. \quad (45)$$

Equations (E1) and (E2) give

$$v = \frac{3mn}{4t}, \quad \ell = -\frac{m(32 + 3h)}{64}. \quad (46)$$

In particular, ℓ is real. Equation (E5) gives

$$k = -7m\frac{n}{\bar{n}}.$$

Equation (E4) shows that k is real, so

$$\sigma := \frac{n}{\bar{n}} \in \{1, -1\}, \quad k = -7m\sigma, \quad n^2 = \sigma|n|^2. \quad (47)$$

After division by appropriate powers of t , equations (E4) and (E3) become

$$3hq^2 + 32q^2 - 448\sigma q - 256 = 0, \quad (48)$$

$$-9hq - 8\sigma h + 112\sigma q^2 - 96q = 0. \quad (49)$$

Eliminating h gives

$$(q - 4)(q + 2)(21q^2 + 42q + 16) = 0 \quad (\sigma = 1), \quad (50)$$

$$(q + 4)(q - 2)(21q^2 - 42q + 16) = 0 \quad (\sigma = -1). \quad (51)$$

The roots $q = -2\sigma$ give $h = -64$ and are impossible. One root of each quadratic gives $h = (-74 - 14\sqrt{105})/3 < 0$. The candidates $q = 4\sigma$ give $h = 32$; then (46) and (E1) imply

$$\frac{|\ell|^2}{t^2} = 64, \quad \frac{|v|^2}{t^2} = 288,$$

and the left side of (E7), divided by t^2 , is 2080, a contradiction. The only remaining possibility is

$$q = -\sigma \left(1 - \frac{\sqrt{105}}{21} \right), \quad h = \frac{-74 + 14\sqrt{105}}{3}. \quad (52)$$

Equation (E6) then gives

$$t^2 = \frac{7(5\sqrt{105} - 33)}{1440}, \quad m^2 = \frac{49\sqrt{105} - 477}{1080}. \quad (53)$$

These three cases exhaust every actual cubic zero, because every such zero satisfies the necessary equations (E1)–(E8).

A.5. Evaluation of the quintic functional. For the actual root $gu_{ij} = (p_i - p_j)/\sqrt{2}$,

$$|z_1|^2 W(z) = \frac{1}{4\sqrt{2}} (a_i - a_j)^3 (\bar{a}_i - \bar{a}_j) (\bar{b}_i - \bar{b}_j). \quad (54)$$

Define

$$\Sigma = \sum_{i < j} X(g)_{ij} (a_i - a_j)^3 (\bar{a}_i - \bar{a}_j) (\bar{b}_i - \bar{b}_j). \quad (55)$$

Then

$$\Lambda_g(Y_H(g)) = \frac{1}{4\sqrt{2}} \text{Im } \Sigma. \quad (56)$$

In wedge notation,

$$X(g) = \frac{i}{2} (\bar{a} \wedge a + 2\bar{b} \wedge b). \quad (57)$$

Modulo an exact matrix, the degree-five edge expression in (55) is

$$\begin{aligned} F \equiv & \bar{b} \wedge (a^3 \bar{a}) + \bar{a} \wedge (a^3 \bar{b}) - (\bar{a} \bar{b}) \wedge a^3 \\ & + 3a \wedge (a^2 \bar{a} \bar{b}) - 3(a \bar{b}) \wedge (a^2 \bar{a}) \\ & - 3(a \bar{a}) \wedge (a^2 \bar{b}) + 3(a \bar{a} \bar{b}) \wedge a^2. \end{aligned} \quad (58)$$

The omitted exact term is $(a^3 \bar{a} \bar{b}) \wedge \mathbf{1}$, which is annihilated by $X(g)$. For the bilinear dot product $x \cdot y = \sum_i x_i y_i$,

$$\sum_{i < j} (u \wedge v)_{ij} (x \wedge y)_{ij} = (u \cdot x)(v \cdot y) - (u \cdot y)(v \cdot x). \quad (59)$$

Every dot product in (59) is five times the coefficient of $\mathbf{1}$ in the corresponding coordinatewise product. Substituting (41), (57), and (58) gives

$$\Sigma = -\frac{i}{80} \mathcal{B}_{\text{raw}}, \quad (60)$$

where

$$\begin{aligned} \mathcal{B}_{\text{raw}} = & -960\bar{k} \ell t + 160 |\ell|^2 m + 320 \ell t^2 - 120 m^3 \\ & - 75m |n|^2 - 800 m t^2 + 100m |v|^2 + 192m - 160 n t \bar{v}. \end{aligned} \quad (61)$$

In all three cases above, ℓ is real. The conjugates of (E4) and (E1) give

$$\bar{k} t = \ell m + 4t^2, \quad n t \bar{v} = \frac{3}{4} m |n|^2. \quad (62)$$

Using (62), followed by (E6) and (E8), reduces (61) to

$$\mathcal{B} = m(180m^2 + 105 |n|^2 - 4400t^2 - 128) - 3520 \ell t^2. \quad (63)$$

The three cases now give

case	$\mathcal{B}$	
I	$160m$	(64)
II	$-80m$	
III	$60m(75 - 7\sqrt{105})$	

Each value is nonzero: in every case $m \neq 0$, and $75 - 7\sqrt{105} > 0$. Since $\mathcal{B}$ is real, (56) and (60) yield

$$\Lambda_g(Y_H(g)) = -\frac{\mathcal{B}}{320\sqrt{2}} \neq 0.$$

This proves the dimension-four assertion of [proposition 3.3](#).

REFERENCES

- [1] X. Jiang, L.-H. Lim, Y. Yao, and Y. Ye, *Statistical ranking and combinatorial Hodge theory*, Math. Program. **127** (2011), 203–244. doi:[10.1007/s10107-010-0419-x](https://doi.org/10.1007/s10107-010-0419-x).
- [2] G. Kuperberg, *Circumscribing constant-width bodies with polytopes*, New York J. Math. **5** (1999), 91–100. <https://nyjm.albany.edu/j/1999/5-6p.pdf>.